



AWS Solution Architect Real-World Scenarios: Practical Q&A for Certification and Interview Preparation Part - 12

Introduction

In this post, we delve into real-world scenarios and advanced use cases to highlight the versatility and efficiency of CloudWatch, and CloudTrail. Whether you're a DevOps Engineer, Database Architect, or Developer, these insights will empower you to harness **CloudWatch, CloudTrail, and AWS troubleshooting tools** capabilities to build scalable, cost-efficient solutions.



The graphic features a dark background with a purple and green banner at the top that reads "AWS SOLUTION ARCHITECT REAL-WORLD SCENARIOS PART 12". On the left is a blue hexagonal badge with the "aws certified" logo and the text "Solutions Architect ASSOCIATE". On the right, a list of topics is shown in purple text: "Monitor systems effectively with CloudWatch alarms and metrics", "Audit and secure environments using CloudTrail logs", "Troubleshooting with AWS Tools", and "Pro Tips for Real-World Scenarios". A circular profile picture of the author and a "Follow" button are also present.

- Monitor systems effectively with CloudWatch alarms and metrics
- Audit and secure environments using CloudTrail logs
- Troubleshooting with AWS Tools
- Pro Tips for Real-World Scenarios

Amazon CloudWatch

Set up CloudWatch Alarms for Monitoring

1. **Scenario:** Your team wants to monitor CPU utilization for EC2 instances and be alerted if it exceeds 80%.

Question: How can you create a CloudWatch alarm for CPU utilization?

Answer:

- In the **CloudWatch Console**, create a new **alarm** using the **CPUUtilization metric** for EC2. Set the threshold to 80% and specify the actions, such as sending an **SNS notification**.
- **Explanation:**



- CloudWatch Alarms allow you to monitor metrics and trigger automated actions when thresholds are exceeded.
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2. **Scenario:** Your organization wants to monitor the number of errors in a specific Lambda function and get an alert if the error count exceeds a threshold.

Question: How can you set up a CloudWatch alarm to monitor Lambda errors?

Answer:

- Use the **AWS Lambda Metrics** for the **Errors** metric in CloudWatch and create an alarm with a threshold for the error count.
 - **Explanation:**
 - Lambda provides built-in metrics for errors, which can be used to set up alarms and notifications for error thresholds.
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3. **Scenario:** Your application requires continuous monitoring of RDS database performance, specifically for disk space usage.

Question: How can you monitor RDS disk space and set an alarm?

Answer:

- Enable the **RDS Enhanced Monitoring** feature and create a CloudWatch alarm on the **FreeStorageSpace** metric.
 - **Explanation:**
 - CloudWatch provides real-time monitoring for RDS metrics, including disk space usage, which helps to prevent potential storage issues.
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Understand CloudWatch Metrics

4. **Scenario:** Your team wants to understand the different CloudWatch metrics for EC2 instances.

Question: What are some common CloudWatch metrics for EC2 instances?

Answer:

- Common metrics include **CPUUtilization**, **DiskReadOps**, **DiskWriteOps**, **NetworkIn**, **NetworkOut**, and **StatusCheckFailed**.
- **Explanation:**
 - These metrics help monitor instance performance, network usage, and disk operations, ensuring the system runs smoothly.



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5. **Scenario:** Your application is running on ECS, and you want to track container performance.

Question: What CloudWatch metrics are relevant for ECS container performance?

Answer:

- Relevant metrics include **CPUUtilization**, **MemoryUtilization**, and **NetworkBytesIn/Out** for each ECS task or service.
- **Explanation:**
 - Monitoring ECS metrics allows you to track resource consumption and optimize container performance.

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6. **Scenario:** Your team needs to monitor an application's latency and response times for API Gateway.

Question: How can you monitor API Gateway performance using CloudWatch?

Answer:

- Use **API Gateway Metrics** such as **Latency**, **4XXError**, **5XXError**, and **Count**.
- **Explanation:**
 - CloudWatch metrics for API Gateway help monitor the performance and error rates of your APIs.

AWS CloudTrail

Review CloudTrail Logs for Auditing

7. **Scenario:** Your organization needs to track who accessed a specific S3 bucket and what actions they performed.

Question: How can you use CloudTrail to review access logs for an S3 bucket?

Answer:

- Ensure **CloudTrail logging** is enabled for S3. Use the **CloudTrail console** to filter logs by the S3 bucket name and access type.
- **Explanation:**
 - CloudTrail logs every API call made to S3, providing detailed information on access and operations.



8. **Scenario:** Your security team wants to monitor all AWS API activities for a specific IAM user.

Question: How can you review CloudTrail logs for a specific IAM user?

Answer:

- Use **CloudTrail Event History** to filter logs by the IAM **user name** and the API actions performed.
 - **Explanation:**
 - CloudTrail captures all user activity, making it easy to track and audit user interactions with AWS services.
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9. **Scenario:** Your compliance team needs to generate reports for all security group modifications.

Question: How can you audit security group changes with CloudTrail?

Answer:

- Filter **CloudTrail logs** by **EC2 SecurityGroup** API actions, such as **AuthorizeSecurityGroupIngress** and **RevokeSecurityGroupIngress**.
 - **Explanation:**
 - CloudTrail logs all changes made to security groups, ensuring that modifications are tracked for auditing and compliance.
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Understand CloudTrail Integration with Other AWS Services

10. **Scenario:** Your team needs to integrate CloudTrail logs with a third-party SIEM tool for real-time monitoring.

Question: How can you export CloudTrail logs to an external system?

Answer:

- Set up **CloudWatch Logs** integration with **CloudTrail** and configure an **AWS Lambda function** to forward logs to the SIEM tool.
 - **Explanation:**
 - CloudWatch Logs can stream CloudTrail logs in real time, and Lambda allows forwarding to third-party tools for further analysis.
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11. **Scenario:** Your team wants to use CloudTrail to monitor and log AWS Config changes.

Question: How does CloudTrail work with AWS Config?



Answer:

- CloudTrail logs all **AWS Config** configuration changes, such as rule evaluations and resource changes.
 - **Explanation:**
 - CloudTrail and AWS Config work together to provide a complete history of resource configuration changes, ensuring security and compliance.
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12. **Scenario:** Your team is using CloudTrail to audit an AWS Lambda function invocation history.

Question: How can you identify Lambda invocations in CloudTrail logs?

Answer:

- Filter **CloudTrail logs** for the **Invoke** action of **Lambda** functions.
 - **Explanation:**
 - CloudTrail logs every invocation of AWS Lambda, enabling auditing of function executions and user access.
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Troubleshooting with AWS Tools

Learn How to Troubleshoot Common Issues Using AWS Tools

13. **Scenario:** Your EC2 instance is not accessible, and you suspect a networking issue.

Question: Which AWS tools can you use to troubleshoot the EC2 instance?

Answer:

- Use **CloudWatch Logs**, **VPC Flow Logs**, and **EC2 Instance System Logs** to analyze network traffic and system-level errors.
 - **Explanation:**
 - CloudWatch and VPC Flow Logs provide insights into networking issues, while EC2 logs help debug system-level problems.
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14. **Scenario:** Your application is experiencing high latency and slow performance.

Question: How can you diagnose performance issues using AWS tools?

Answer:

- Use **CloudWatch Metrics** to monitor resource utilization (e.g., CPU, memory) and **CloudWatch Alarms** to alert on performance thresholds.



- **Explanation:**
 - CloudWatch helps you track performance and set alarms to notify you of performance issues in real time.
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15. **Scenario:** Your Lambda function is failing intermittently due to resource limitations.

Question: How can you troubleshoot Lambda execution issues?

Answer:

- Use **CloudWatch Logs** to inspect Lambda function logs and **CloudWatch Metrics** to analyze execution duration and resource usage.
 - **Explanation:**
 - Logs capture detailed error messages, and metrics help determine if resource limits (e.g., memory, timeout) are being exceeded.
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Use AWS Config for Resource Inventory

16. **Scenario:** Your organization needs to ensure all EC2 instances are in a specific security group.

Question: How can you check compliance using AWS Config?

Answer:

- Set up an **AWS Config rule** to evaluate whether EC2 instances are associated with the correct security group.
 - **Explanation:**
 - AWS Config enables continuous monitoring of resource configurations, allowing you to automate compliance checks for security standards.
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17. **Scenario:** Your team needs an inventory of all running EC2 instances and their configurations.

Question: How can you gather an inventory of EC2 instances using AWS tools?

Answer:

- Use **AWS Config** to track EC2 instances and their configurations or use **AWS Systems Manager Inventory** for detailed resource collection.
- **Explanation:**
 - AWS Config provides a detailed history of configurations, while Systems Manager can gather current state data for inventory management.



18. **Scenario:** Your organization wants to ensure that only compliant AWS resources are deployed.

Question: How can you enforce compliance on AWS resources?

Answer:

- Use **AWS Config Rules** and **AWS Organizations** with **Service Control Policies (SCPs)** to enforce resource compliance.
 - **Explanation:**
 - AWS Config Rules automate compliance checks, and SCPs allow central governance of policies across AWS accounts.
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Set up CloudWatch Alarms for Monitoring

19. **Scenario:** Your team needs to monitor and receive alerts when an EC2 instance's disk usage exceeds 85%.

Question: How can you set up a CloudWatch alarm for EC2 disk usage?

Answer:

- Create a CloudWatch alarm using the **DiskSpaceUtilization** metric from **CloudWatch Agent** installed on the EC2 instance. Set the threshold to 85%.
 - **Explanation:**
 - CloudWatch Agent allows you to monitor custom metrics like disk usage, and setting an alarm helps prevent disk-related performance issues.
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20. **Scenario:** Your team needs to track and receive alerts for AWS Lambda execution duration exceeding a threshold.

Question: How can you create an alarm for high Lambda execution time?

Answer:

- In CloudWatch, select the **Duration** metric for your Lambda function and create an alarm for values exceeding your threshold.
 - **Explanation:**
 - Monitoring Lambda execution time ensures that functions perform within acceptable time limits and helps identify bottlenecks.
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21. **Scenario:** Your organization needs to be alerted when EC2 instances experience status check failures.

Question: How can you set up an alarm for EC2 status check failures?

Answer:

- Create an alarm using the **StatusCheckFailed** metric for EC2 instances and set the threshold to 1 for failure detection.
 - **Explanation:**
 - CloudWatch will trigger an alarm when the EC2 instance's status check fails, enabling rapid issue resolution.
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Understand CloudWatch Metrics

22. **Scenario:** Your team wants to track memory usage for an EC2 instance, but the default CloudWatch metrics do not include memory metrics.

Question: How can you monitor EC2 memory usage in CloudWatch?

Answer:

- Install and configure **CloudWatch Agent** on the EC2 instance to collect memory metrics and send them to CloudWatch.
 - **Explanation:**
 - CloudWatch Agent enables the collection of custom metrics, such as memory usage, which is not included by default for EC2 instances.
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23. **Scenario:** Your team needs to monitor RDS instance performance, specifically read and write IOPS.

Question: Which CloudWatch metrics should you monitor for RDS IOPS?

Answer:

- Monitor the **ReadIOPS** and **WriteIOPS** metrics under the RDS instance in CloudWatch to track I/O operations.
 - **Explanation:**
 - These metrics provide insights into the read and write operations performed on the RDS instance, helping identify performance bottlenecks.
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24. **Scenario:** Your organization wants to ensure that your S3 bucket's storage is not growing unexpectedly.



Question: How can you track S3 storage usage in CloudWatch?

Answer:

- Use the **S3 Storage Metrics** in CloudWatch, such as **BucketSizeBytes**, to monitor storage usage over time.
 - **Explanation:**
 - These metrics help track storage growth, allowing you to set up alarms to detect unexpected increases in data volume.
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Review CloudTrail Logs for Auditing

25. **Scenario:** Your team needs to audit changes made to IAM roles for compliance purposes.

Question: How can you track changes to IAM roles in CloudTrail?

Answer:

- Filter **CloudTrail logs** by **IAM role** actions such as **CreateRole**, **AttachRolePolicy**, or **UpdateAssumeRolePolicy**.
 - **Explanation:**
 - CloudTrail records every IAM action, making it easy to track role creation, policy attachments, and modifications.
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26. **Scenario:** Your organization needs to identify any unauthorized API calls made within a specific AWS region.

Question: How can you filter CloudTrail logs for API calls in a specific region?

Answer:

- Use **CloudTrail Event History** and filter logs by **region** and **API action** to identify any unauthorized activities.
 - **Explanation:**
 - Filtering by region helps track activities specific to a geographical location, while analyzing unauthorized API calls enhances security audits.
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27. **Scenario:** Your security team needs to investigate potential unauthorized access to an S3 bucket.

Question: How can you investigate unauthorized access to an S3 bucket using CloudTrail logs?



Answer:

- Filter CloudTrail logs for S3 **GetObject** and **PutObject** actions, and analyze the **source IP** and **user identity** associated with each access request.
 - **Explanation:**
 - CloudTrail logs contain detailed access information, including the identity of the requester and the action performed, aiding in identifying unauthorized access.
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Understand CloudTrail Integration with Other AWS Services

28. **Scenario:** Your organization wants to set up automatic notifications when CloudTrail logs detect changes to an S3 bucket.

Question: How can you configure notifications for changes in CloudTrail logs?

Answer:

- Use **CloudWatch Logs** to create a subscription for CloudTrail logs and trigger an **SNS topic** to send notifications for specific S3 bucket changes.
 - **Explanation:**
 - CloudWatch Logs can filter CloudTrail logs for specific actions, and SNS ensures real-time alerts for critical events like unauthorized access to S3.
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29. **Scenario:** Your compliance team needs to ensure that CloudTrail logs are stored securely for auditing purposes.

Question: How can you secure CloudTrail logs?

Answer:

- Store CloudTrail logs in **S3** with **server-side encryption (SSE)** using **KMS** and enable **access logging** to track who accesses the logs.
 - **Explanation:**
 - Encrypting logs and enabling access logging helps ensure secure and compliant storage of audit logs.
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30. **Scenario:** Your organization needs to analyze CloudTrail logs to detect potential misconfigurations in IAM policies.

Question: How can CloudTrail help detect misconfigurations in IAM policies?

Answer:



- Review **IAM policy-related events** in CloudTrail, such as **AttachUserPolicy** or **DetachUserPolicy**, and use **CloudWatch Insights** to query logs for unusual access patterns.
- **Explanation:**
 - CloudTrail logs provide detailed information about IAM policy changes, which can be cross-checked with expected configurations for anomalies.

Understanding how to leverage AWS tools and features will enhance your capabilities, support certification preparation, and boost confidence in real-world problem-solving for DevOps, cloud engineering, and SRE roles. In the up-coming parts, we will discussion on more such practical challenges along with steps for the different AWS based scenarios. So, stay tuned for the and follow @Prasad Suman Mohan for more such posts.

